

Orientation Discrimination Task

PSY310: Lab in Psychology

Lab Report

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GitHub Link-

<https://github.com/HetanshJalan/Psychology-Experiments>

Introduction

Orientation discrimination refers to the capacity to discriminate orientation of visual patterns. Grating is very useful in psychophysics because sinusoidal gratings can be oriented, spatially modulated, contrasted, and displayed accurately and precisely. Studies of orientation discrimination reveal that it is a function of stimulus properties and sensitivity of the visual system to orientation difference (Beaudot & Mullen, 2006).

Using computer-based psychophysics, stimulus duration can be controlled and brief visual stimuli can be presented with responses and reaction times being recorded. PsychoPy is intended to facilitate behavioural experiments that require very precise presentation of stimuli and collection of responses (Peirce et al., 2019). In the present experiment, a sinusoidal grating was presented for one minute, at one of five different orientations, using PsychoPy, and a binary keyboard response was recorded.

The orientation discrimination task is described in this submission, along with an adaptive staircase version, which is described in the reference report provided with this submission. Later observations are typically placed efficiently around a performance threshold in transformed up-down procedures in psychophysics (Levitt, 1971). The currently available .psyexp does not support adaptive staircase. Rather, it employs a 10-trial randomized TrialHandler and produces one of two stimuli: a vertical stimulus (0°) or a non-zero tilt ranging from 1° to 5° either ipsilateral or contralateral to the optic axis. Hence, the present report does not claim to perform threshold estimation staircase but rather as an orientation-discrimination pilot.

Aim- To investigate initial performance at discriminating a vertical grating from a tilted grating, accuracy and reaction time over the tilt magnitudes sampled in the pilot run.

Hypothesis- The accuracy would be better with larger magnitudes of tilt than with smaller non-zero magnitudes of tilt as there would be more differences to be discerned with a larger magnitude of tilt.

Method

Participant

The experiment was conducted by a single 20-year-old male undergraduate student, who also was the student researcher. The run was performed as a pilot and the data collected was stored under that name "pilot" in the recorded data. No outside recruitment was needed. No additional procedural claim is made here because the materials provided do not provide for a separate informed consent procedure.

Materials

The experiment was run using PsychoPy version 2026.2.3. It used a full-screen display with dimensions 1024×768 measured in PsychoPy height units. During the run the frame rate came out to about 60 frames per second. The computer used for the experiment had a generation Intel Core i3 processor, NVIDIA GeForce MX350 graphics.

The visual stimulus was a sinusoidal grating covered by a Gaussian mask. In the PsychoPy file the grating was sized at (0.2, 0.2) in PsychoPy height units. It had a contrast of 0.3, a frequency of 5 cycles, per stimulus unit and its orientation was controlled by the variable 'tilt'. A white fixation cross was shown in front of the grating to help participants focus. Participants responded using the keyboard keys labeled "up" and "down".



Figure 1. Example of the orientation stimulus supplied with the experiment.

Experiment parameters

Parameter	Setting
Software	PsychoPy 2026.2.3
Trials	10; randomized TrialHandler
Tilt generation	0° or $\pm 1^\circ$ to $\pm 5^\circ$
Fixation	White cross, 1,000 ms
Grating duration	300 ms
Grating size	(0.2, 0.2) PsychoPy height units
Spatial frequency	5 cycles per stimulus unit
Contrast	0.3
Mask	Gaussian
Response keys	"up" and "down"

Correct response	"up" for 0°; "down" for any non-zero tilt
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Table 1. Main experiment parameters obtained from the supplied PsychoPy experiment file.

Procedure

The participant was seated at the computer and asked to answer the orientation stimulus. All trials started with a white square fixation cross displayed at the center of the screen for 1000ms. A sinusoidal grating was then presented for 300ms. The grating was either set vertical (0°) or tilted to the left or right by 1°–5°; the sign and amount of the tilt were determined randomly by the experiment code. The response keyboard component became active with the grating and accepted either the "up" or "down" key. For a vertical grating, a programmed response mapping had the answer "up"; for any non-zero tilt, it had the answer "down". A programmed response mapping was correct for a vertical grating, and it was correct for any non-zero tilt it was not. The experiment was ended with the answer and the next random experiment commenced.

The provided TrialHandler was configured to perform 10 trials in a random order. No adaptive adjustment of tilt based on previous responses was found in the current .psyexp file, and therefore there were no staircase reversals and there was no threshold-convergence rule in this run.

Data treatment

Trial number, tilt, response, correctness, reaction time were checked in the CSV output. Reaction times in seconds were converted to milliseconds. Absolute tilt magnitude was determined as the absolute value of the signed tilt in order to combine left and right oriented stimuli. The percentage of correct answers was used as a measure of accuracy. Descriptive summaries were computed overall, and for each magnitude of absolute tilts observed. Since only one participant is included and 10 trials only, no inferential statistical test was considered a meaningful estimate of population performance.

Results

There were 10 trials overall. The participant correctly answered 4 out of the 10 questions, for an overall accuracy of 40%. Mean reaction time was 652.5 ms (SD = 223.0 ms), with a median of 609.7 ms.

There were 2 vertical trials (0°) and 8 tilted trials. Accuracy was 100.0% for vertical trials (2/2) and 25.0% for non-zero tilted trials (2/8). The participant gave the "up" response 8 times out of 10, which means that the response was more frequently given for the vertical response than the tilted one.

Absolute tilt (°)	Trials	Correct	Accuracy	Mean RT (ms)
0	2	2	100.0%	944.2

1	1	0	0.0%	467.2
2	6	1	16.7%	603.1
3	1	1	100.0%	550.5

Table 2. Observed accuracy and mean reaction time at each absolute tilt magnitude.

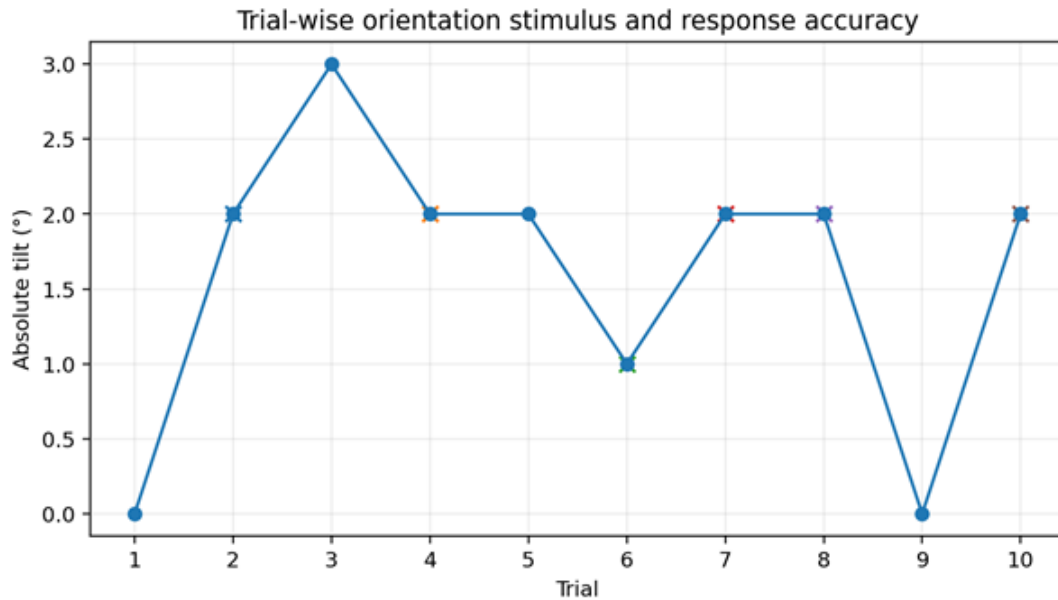


Figure 2. Trial-by-trial absolute tilt magnitude. Crosses indicate incorrect responses; circles indicate the recorded trial points.

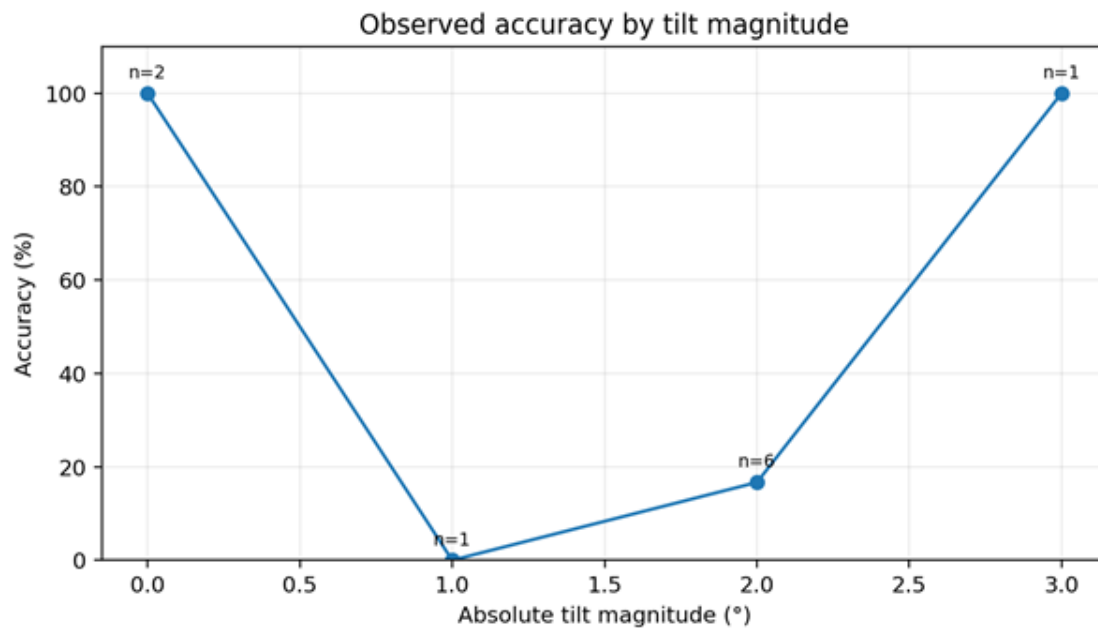


Figure 3. Observed accuracy by absolute tilt magnitude. The small number of observations at each level should be considered when interpreting the percentages.

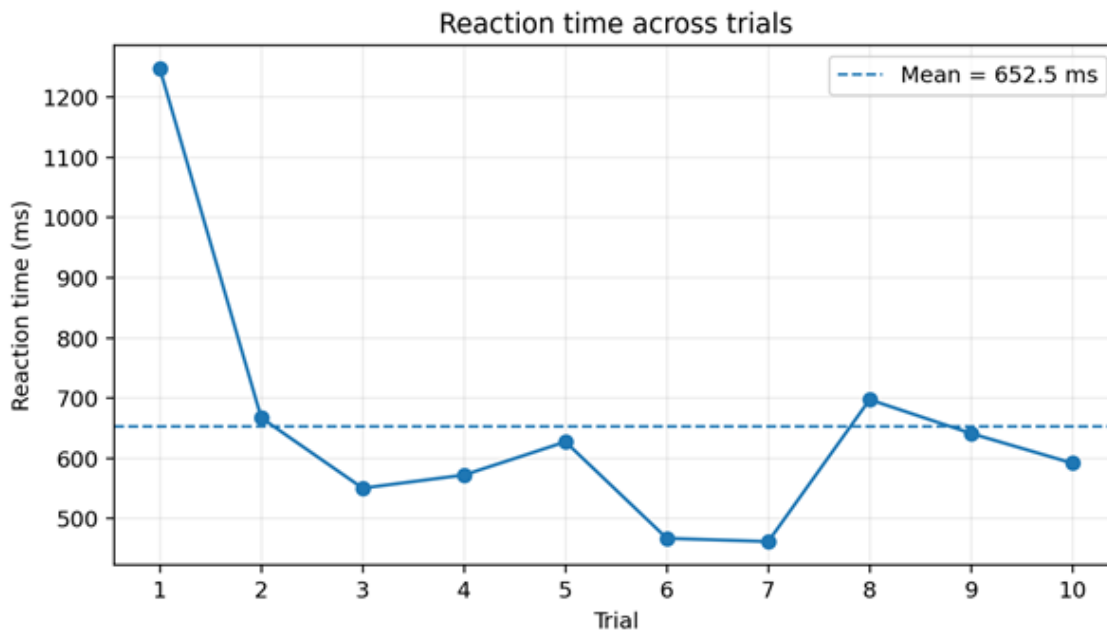


Figure 4. Reaction time across the 10 trials. The dashed line indicates the overall mean reaction time.

Discussion

The accuracy of the pilot was 40.0% (4/10), and there was no evidence that the hypothesis that larger differences in orientation were always easier to discriminate was supported. The participant was correct for 2/8 non-zero tilt trials and was correct for 2/2 trials on 0° trials. With the individual magnitudes, 0% of the trials were successful at 1°, 16.7% at 2° and 100% at 3°; but only 2 trials and one trial were made at 0° and 3°, respectively. Therefore, the apparent tilt-dependent difference in the psychometric function should not be treated as a stable psychometric function.

One of the key aspects of the data was the response pattern. The participant performed on 8 of the 10 trials with a press of "up", of which 6 were tilted trials. This tendency to respond "down" led to six errors to the tilted stimuli, since the correct response was "down" for all stimuli other than "up". The low overall accuracy is thus most likely a response bias to the vertical category as well as a difficulty of the small tilts. At the present stage, these are not distinguishable in the available data with 10 trials.

The experiment nevertheless provides a simple demonstration of the principles of orientation-discrimination psychophysics. Small orientational changes can be used to measure visual orientation sensitivity (Beaudot & Mullen, 2006) and controlled grating stimuli can be used to carefully control the physical properties of the stimuli. Timing and response collection is also reproducible (Peirce et al., 2019) using the use of PsychoPy.

A key methodological issue here is that it is important to not refer to the current pilot as an adaptive staircase experiment. The supplied reference report was written with a 3 down/1 up staircase, while the current .psyexp file is based on a randomized 10 trials TrialHandler and will not make changes to stimulus intensity based on previous behavior. In a "true" up/down, the level of the stimulus is adjusted on the basis of the response history so that observations can focus on a target performance level (Levitt, 1971). The present code does not include that mechanism; therefore, from these 10 trials a threshold or a reversal-based estimate will not be possible.

The drawbacks are that the study is conducted on a single participant, the number of trials is very small, and the tilt magnitudes are not evenly distributed. Even though the code could produce tilts of up to 5° the pilot was only sampling a section of the programmed ones, the largest observed being 3° in this particular run. If the goal of the follow-up is to estimate either an orientation-discrimination threshold, then it should employ a conventional psychometric design or a properly implemented adaptive staircase, with many more trials and with equal numbers of observations at each tilt, and with responses mapped to the tilt (or counterbalancing if applicable). Within-participant reliability could also be evaluated if the participants would be run again.

Conclusion

The 40.0% accuracy obtained in the orientation-discrimination pilot, obtained over 10 trials, with a mean reaction time of 652.5 ms, was perfect on the two vertical trials, and far lower on non-zero tilted trials, in part due to the participant's choice of the vertical response in most of the trials. The results, accordingly, cannot be used to estimate an orientation threshold and the accuracy-by-tilt relationship is not clear. The experiment was successfully presented in PsychoPy, and recorded a response, however there was a significant need for much more trials and a more solid psychophysical design to obtain a meaningful threshold or psychometric-function estimate.

References

- Beaudot, W. H., & Mullen, K. T. (2006). Orientation discrimination in human vision: Psychophysics and modeling. *Vision research*, 46(1-2), 26-46.
- Levitt, H. C. C. H. (1971). Transformed up-down methods in psychoacoustics. *The Journal of the Acoustical society of America*, 49(2B), 467-477.
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., ... & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior research methods*, 51(1), 195-203.

Appendix : Trial data

The table below reproduces the main trial-level variables from the recorded CSV output. Reaction time is shown in milliseconds; 1 = correct and 0 = incorrect.

Trial	Tilt (°)	Absolute tilt (°)	Response	Correct	RT (ms)
1	0	0	up	1	1247.1
2	-2	2	up	0	667.3
3	3	3	down	1	550.5
4	-2	2	up	0	572.3
5	2	2	down	1	627.4
6	1	1	up	0	467.2
7	-2	2	up	0	461.8
8	-2	2	up	0	697.5
9	0	0	up	1	641.4
10	2	2	up	0	592.0